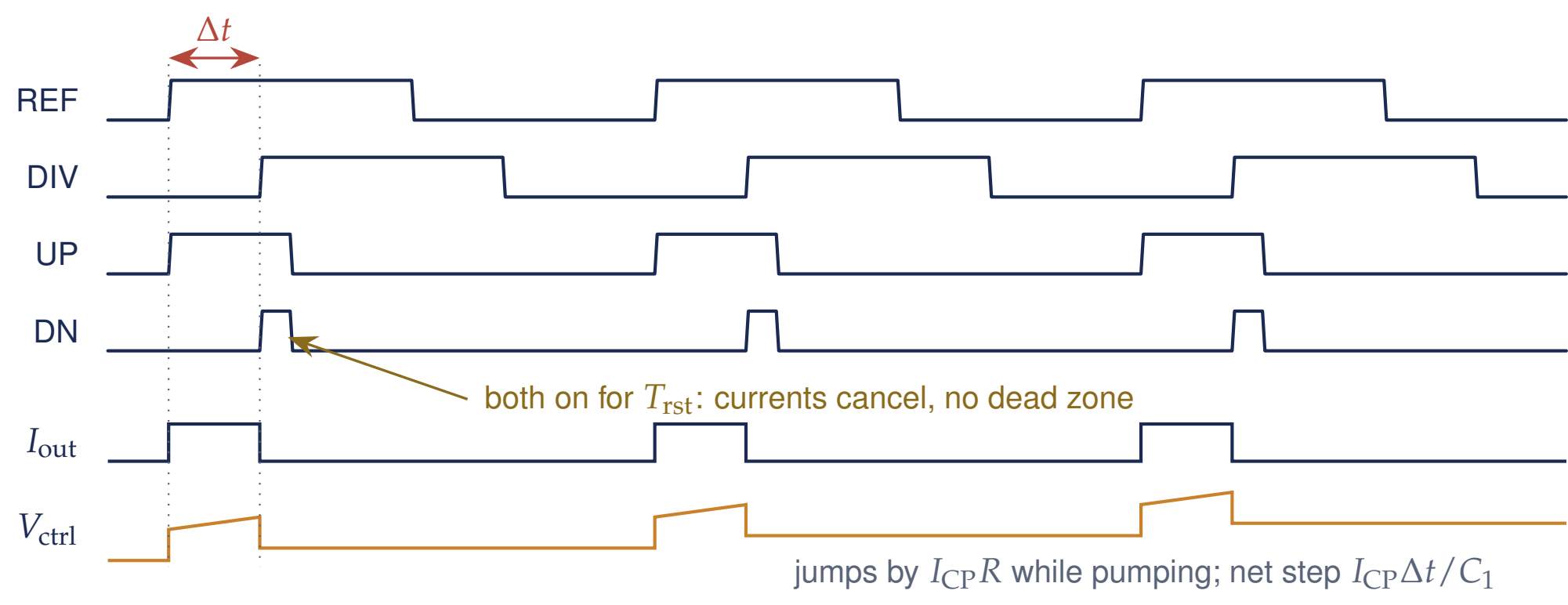
Δt
REF
DIV
UP
DN
both on for T_{rst}: currents cancel, no dead zone
I_{out}
V_{ctrl}
jumps by $I_{\mathrm{CP}}R$ while pumping; net step $I_{\mathrm{CP}}\Delta t / C_1$